

Ishaan Jain

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PERSONAL SUMMARY

Versatile Engineering Physicist bridging the gap between theoretical R&D and hands-on hardware implementation. With my master's thesis at IMEC, I possess expertise in semiconductor device modeling (TCAD), Python-based data automation, analytical physics, and basic material characterization. Alongside my simulation capabilities, I bring extensive hands-on experience in high-voltage system manufacturing, custom PCB debugging, and opto-mechanical instrument design, manufacturing and assembly. With a strong theoretical foundation in process design and material characterization, paired with highly adaptable electrical-mechanical skills, I am equipped to drive value across R&D, process simulation, equipment engineering, and hardware testing roles.

EDUCATION

KU LEUVEN

Leuven, Belgium

Master of Science

2024 – 2026

Nanoscience, Nanotechnology and Nanoengineering; specialization in Nanodevices and Circuits

Cumulative GPA: Cum Laude (70%, KU Leuven scale);

Thesis Title: Modeling MOL (middle-of-line) parasitics for advanced CFET nodes

INDIAN INSTITUTE OF TECHNOLOGY DELHI

New Delhi, India

Bachelor of Technology

2020 – 2024

Engineering Physics; Minor in Electrical Engineering

Cumulative GPA: First Class (75%);

Thesis Title: Design and Simulation of Single Photon Avalanche Diodes (SPADs) for IR Camera sensors

WORK EXPERIENCE

INTERUNIVERSITY MICROELECTRONICS CENTRE (IMEC)

Leuven, Belgium

Master Thesis Student Researcher

Jul 2025 – Aug 2026 (completed)

- Executed Device Technology Co-Optimization (DTCO) for advanced 3D CFET nodes in Synopsys TCAD, fully parametrizing the device across 43 distinct geometric/design parameters to enable systematic multi-variable design-space exploration.
- Authored Python automation to extract SPICE-compatible parasitic capacitance models from TCAD simulation data and applied the Kron reduction network method to reduce parasitic networks to input/output/node capacitances, achieving under 10% error; built a custom TCAD log-file visualizer that substantially accelerated data rendering versus the native simulation software.
- Generated SPICE-compatible netlists structured for Process Design Kit (PDK) integration, validating analytical CFET parasitic capacitance models against simulation results.

TATA INSTITUTE OF FUNDAMENTAL RESEARCH (TIFR)

Mumbai, India

Opto-Mechanical Instrument Engineering Intern

Jun 2023 – Aug 2023

- Designed a highly precise opto-mechanical robotic arm utilizing SolidWorks CAD, employing Design for Manufacturability and Assembly (DFMA) principles to mitigate manufacturing defects and account for strict assembly tolerances.
- Minimized structural deflections and mechanical vibrations utilizing Ansys Workbench FEA, and validated optical alignments using Zemax optical solutions to achieve a strict 16-micrometer and 5-microradian assembly precision.
- Engineered the robotic system to successfully redirect a selected 13mm² optical field from a massive 10m telescope aperture directly into advanced spectroscopy equipment. Whole thing was assembled in a clean room to reduce contamination in spectroscopy array.

PUBLICATION

Mechanical design of the pick-up arm prototype for TIFR-ARIES Multi-Object Optical to Near Infrared Spectrometer (TA-MOONS)

Proc. SPIE 13096, Ground-based and Airborne Instrumentation for Astronomy X (July 2024) | Role: Contributing author (not first author).

PROJECTS

SIMULATION AND OPTIMIZATION OF OPTICAL NEMS-BASED XOR GATE

Leuven, Belgium

Course Project: Project & Design, IMEC

Feb 2025 - Jun 2025

- Optimized the dimensional and electrical parameters of a MEMS device using COMSOL Multiphysics and analytical MATLAB models.

OPERATIONAL TELESCOPIC AMPLIFIER DESIGN

Leuven, Belgium

Course: Design and Implementation of Analogue Circuits, MICAS

Sep 2024 - Nov 2024

- Achieved 87.65 dB gain and 1.23 MHz gain-bandwidth product within power (1mW) and area constraints, by designing an OTA (180nm node) in Cadence Virtuoso, analyzing fabricated (tape-out) microchips to confirm sub-1% deviation.

DESIGN AND SIMULATION OF SINGLE PHOTON AVALANCHE DEVICE (SPAD)

Delhi, India

Bachelor Thesis Project

Aug 2023 – May 2024

- Developed structural optimization models for advanced infrared camera sensors, utilizing Silvaco Atlas TCAD to systematically evaluate Planar and Epitaxial Single Photon Avalanche Diodes (SPADs).
- Achieved a targeted 30% Photon Detection Efficiency (PDE) at 800nm and a -18.1V breakdown voltage by iterating structural parameters against industry baseline metrics.

XLR-20 Electric Race car — Formula Student (AXLR8R Club)

Delhi, India

Accumulator Engineer

May 2021 – Jun 2022

- Designed and manufactured a 400V Lithium-Ion pouch cell battery module with a custom BMS design and balancing algorithm, including 2 custom PCBs — a per-cell sensing board replicated 8x to monitor all 192 cells, plus a central data-aggregation board.
- Delivered complex electro-mechanical systems under intense financial and project deadlines, successfully managing a ~10 million INR equipment budget, by coordinating hardware activities and strict timelines with vendors.
- Troubleshoot critical electrical issues, conducting rigorous root-cause analysis with oscilloscopes and Source-Measure Units (SMUs) to achieve zero electrical failures across 50+ hours of continuous capacity testing of multiple self-designed PCBs.

SKILLS

Simulation & Advanced Modeling: Synopsys TCAD, Silvaco Atlas, COMSOL Multiphysics, Ansys Workbench (FEA), Zemax (OpticStudio), Cadence Virtuoso, Process Sensitivity Analysis, Theoretical Process Recipe Design.

Programming & Automation: Python (NumPy, Pandas, Matplotlib), Automated Measurement Recipes, MATLAB/Simulink, C/C++.

Hardware, Electronics & Troubleshooting: High-Voltage Systems (400V), Custom PCB Debugging, Oscilloscopes, Source-Measure Units (SMUs), Opto-Mechanical Assembly & Tuning.

Lab Operations & Metrology: Automated Atomic Force Microscopy (AFM) Profiling, Physical Characterization, 2-/4-Probe Measurements, Hall Effect Characterization, Lab Sample/Wafer Handling, Cleanroom Protocol Exposure, Electrostatic Discharge (ESD) Safety.

Mechanical Design & Manufacturing: SolidWorks (CAD), Geometric Dimensioning and Tolerancing (GD&T), 3D Printing (Additive Manufacturing), Laser Cutting, Sheet Metal Fabrication, CNC & Manual Machining.

Languages: English (Fluent, IELTS 7.5), Dutch (A2), Hindi (Native), Japanese (Beginner).

RELEVANT COURSEWORK

Material Physics & Technology for Nanoelectronics, Physical Material Characterization, Material Characterization Techniques Microscopy & Diffraction, Semiconductor Devices, Technology of Integrated Systems, Solid State Physics, Reliability & Yield, Design and Implementation of Analogue Circuits, Design of Digital ICs.

EXTRACURRICULAR

- **Sakura Science Exchange Fellow** (*Japan Science and Technology Agency, 2024*): Selected for a fully sponsored, competitive exchange program for advanced research in Japan, with exposure to top Japanese universities and R&D centers.
- **National STEM Honors:** IIT JEE Advanced AIR 1756 (top ~0.1% nationally); KVPY Fellow (Govt. of India Dept. of Science & Technology); Top 0.1% nationally in Physics & Astronomy Olympiads to go for international Olympiads. IEEE Student Member (current affiliation).
- **Head, Physics and Astronomy Society** (*IIT Delhi, 2021–2023*): Organized large-scale astronomical observation events and founded the university's first student-led rocketry program.
- **Student Leadership & Community:** Elected Hostel Representative for 600 students, managing welfare budgets and mentoring incoming first-years; volunteered tutoring underprivileged students in core STEM subjects.

REFERENCES

"Available upon request"